

Society for Research in Psychopathology: Abstract Submission

Submission Type: Poster/Oral Presentation

Title of Abstract: Prenatal Complications Predict Thought Disorders in the Adolescent Brain Cognitive Development Study.

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Primary Category: Schizophrenia/Psychosis

Secondary Category: Neuroimaging

Current Level of Training: Post-baccalaureate research assistant

Background: Individuals with schizophrenia have higher rates of prenatal complications (PCs), with each type reflecting distinct pathways of risk for psychosis. Specific insults (e.g., hypoxia, prenatal infection, maternal stress) predict earlier psychosis onset and increased symptom severity, but also cognitive difficulties through effects on areas like the hippocampus. Thus, the present study aims to explore associations between PC loading and type on psychotic-like experiences (PLEs) and thought disorders through hippocampal pathways.

Methods: Baseline data from 9,704 subjects (singleton birth; ages 9-10) in the Adolescent Brain Cognitive Development Study were assessed on PC burden and type, PLE severity, thought problems, and hippocampal volumes. Linear regression models were used to model potential relationships between PCs, PLEs, thought problems, and hippocampal abnormalities.

Results: Total number of PCs was significantly related to PLE severity ($t=5.380$, $p < .001$) and thought problems ($t=7.306$, $p<.001$). There was specificity in the relevance of PC types to Hypoxia ($t=3.08$, $p=.002$) and Maternal Stress ($t=2.613$, $p=.009$), with these factors significantly contributing to the model, while Prenatal Infection and Substance Exposure did not contribute as expected ($p's>.07$). Additional analyses will be presented regarding the potential moderating role of hippocampal neurodevelopment in psychosis risk.

Conclusions: The present study provides evidence that PC burden and type, specifically hypoxia and maternal stress, may predict an individual's likelihood of developing distressing PLEs and thought problems during the transition to adolescence. PCs continue to be one of the earliest markers of at-risk individuals and can aid in the provision of earlier, more targeted interventions.

Keywords: Hippocampal Volumes, Obstetric Complications, Adolescent Brain Cognitive Development (ABCD) Study, Neurodevelopment, Psychotic-like experiences (PLEs)

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